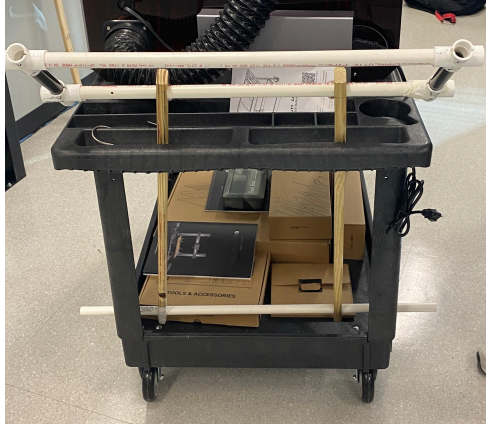


Element F

The problem at hand revolves around trying to create a product that is both ergonomically sound and functional at the same time. In terms of functionality, the product must meet certain criteria provided by the client. The different criteria include adjustability, ergonomic ability, budget, removability, and durability along with universal use, and doesn't impede any storage space. The problem must be able to satisfy the client's need to push the cart at a height where he doesn't have to overextend causing pain in his lower back.

The design focuses on ergonomics and functionality for the most effectiveness. The design incorporates a series of materials including wood, superglue, rubber, and some PVC. The biggest drawback is that the budget is extremely limited and we are only allocated \$30 to fund our project. This makes this project that much harder under these limitations because it doesn't give us much to work with and forces us to spend the money very cautiously on only things that we need. In terms of materials we are able to use anything possible, but keep the materials as cheap to help us down the line. There aren't any specific requirements as to how this product should be designed or what they should be built with.



Some of the extra-functional evaluations of our design include the cost-effectiveness, scalability, user experience, and sustainability of our design. In terms of budgeting, we have a fairly positive outlook with a cheap effective design. Our design should cost around \$12.59 (\$2.61 for all the wood, \$7 for PVC elbow fittings, and \$2.98 for 76 inches of $\frac{1}{2}$ " PVC) assuming we can get the parts to the exact specs that are desired. Assuming we buy the materials in full from [Home Depot](https://www.homedepot.com/) we would be spending about \$19.39 (\$7.68 for an 8ft plank, \$4.71 for a 10ft pipe, and \$7 for the PVC elbow fittings) for all the materials plus excess materials. This design is extremely cost-effective and would save money for the future to evolve the design however necessary after further testing. The biggest advantage of our design is the fact that it is cheap enough to be able to expand past its current state. With the right investors, this product could rise from small-scale production to large-scale production. The idea is that this product is extremely scalable and was designed to be produced at a smaller scale until the option was given to evolve this design into a better, more efficient product that suits all the client's needs. The issue currently is that the funding is at a bare minimum and we are fairly limited as to what we are allowed to use since the budget is so low. The user experience is yet another function

that makes our product stand out from the rest. Within our design, we have several user-friendly functions that allow for easy use. The product comes with user-friendly details breaking down how to use the product. The last thing is the sustainability of the design. We specifically designed this product to last the designated amount of time. The materials for our design were specifically chosen to last long term because this criterion was one of higher priority. We utilized wood we found and evaluated it to be sufficient enough to withstand the pressure of being wedged between the handle and the bottom of the cart. The cart handle extension uses lightweight, durable materials such as PVC and wood, which are affordable and commonly available. This helps keep production costs low without sacrificing quality. Whilst this design is durable and functional, the design is not necessarily environmentally friendly. This poses an issue because PVC is often toxic and can be harmful to children's health. Considering the pros of the design significantly outweigh the cons, the next step is being able to resolve the issues that are present in the design currently. The design can be improved with the help of investors who can potentially introduce other ideas to help resolve some of the current issues at hand in the design.